PROPOSAL PRACTICE CHALLENGE

PlaySafe

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1.Introduction and context

- **What?:** The aim of this project is to develop a machine learning binary classification model that estimates the probability of a football player in LaLiga suffering at least one injury during the upcoming season. Instead of only giving a yes/no answer, the model will provide a continuous injury risk score. For example, Player A could have an estimated injury risk of 72%.

- **Why?**: Player injuries can have a big impact on team performance, tactical planning, and financial investments in professional football. By identifying players who may have a higher risk of injury before the start of the season, clubs can make better decisions about player rotation, training load, and injury prevention.

- **Who?**
  Primary stakeholders: Sports scientists, physical trainers, and medical staff at LaLiga clubs. They can use the risk predictions to adapt training and physical preparation to individual players.

  Secondary stakeholders: Technical directors and analysts who need to evaluate squad depth and the possible injury risk of players before transfer windows or when planning a new season.

2.Domain understanding

- **Business/Domain Context:** In professional football, such as LaLiga, injuries can be influenced by different factors, including physical workload, player characteristics, the number of matches played in a short period, and previous injuries.
  Traditionally, injury prevention has mainly depended on medical knowledge, clinical experience, and physical data such as GPS measurements. Machine learning can provide another approach by analysing historical data and identifying relationships between different factors that may be difficult to detect manually.

- **Problem Statement:** The main problem is to predict whether a player will suffer at least one injury in a future season without using information from that future season. This is important because using future information could create data leakage and make the model look more accurate than it really is.
  Another challenge is that there may be more players without injuries than players who suffer injuries, creating a class imbalance problem. The model also needs to provide reliable probability scores instead of only giving a simple injured/not injured classification.

- **Project Objectives:**

1- Build a reliable machine learning pipeline that avoids data leakage by using only historical information available before the start of the target season.
2-Evaluate the model using a time-based validation approach, training the model on previous seasons and testing it on a later season. The objective is to perform better than a simple random baseline.
3-Produce reliable probability estimates. For example, if the model predicts a 70% injury probability for a group of players, the actual injury rate for similar predictions should be reasonably close to 70%.

3.Model requirements

3.1Analytic Approach

Problem Type: Supervised binary classification with probability estimation.

Algorithms Evaluated

1. Logistic Regression

Logistic Regression will be used as the baseline model. It is a simple and interpretable model that can directly provide a probability of injury for each player.

2. Random Forest

Random Forest will be used as a second model to compare its performance with Logistic Regression. It can capture more complex relationships between variables without requiring too much manual configuration.

These two models provide a good balance between simplicity and performance. More complex models such as XGBoost or neural networks will not be used because they would add complexity without being necessary for the main objective of this project.

These models will be tested because they can capture non-linear relationships and interactions between different variables.

Validation Strategy: A time-based split will be used to make the evaluation more realistic. The model will be trained using previous seasons and then tested on a later season.

For example, the model could be trained using several previous seasons and then tested on the following season. This is important because, in a real situation, we would only have information from previous seasons when predicting future injuries.

This approach also helps to avoid data leakage and look-ahead bias.

Evaluation Metrics:

To evaluate the models, four common metrics will be used:

- Accuracy: The percentage of predictions that are correct.
- Precision: The percentage of players predicted as injured who actually suffer an injury.
- Recall: The percentage of players who actually suffer an injury that the model correctly identifies.
- F1-Score: A metric that combines Precision and Recall into a single value.

Recall will be especially important for this project because the objective is to identify as many players at risk of injury as possible. However, Precision and F1-Score will also be considered to have a more complete view of the model's performance.

3.2 Data Requirements

Data Source:
The project will use the laliga_players.csv dataset from the *football-injury-scraper* GitHub repository. The dataset contains information about LaLiga players, their playing time, seasons, and recorded injuries.

Target Variable:
The target variable will be has_injury_next_season.

- 0 = No recorded injury in the target season.
- 1 = At least one recorded injury in the target season.

The target will be created using the historical season and injury information.

Features:
The dataset contains several types of information that can be used as potential predictors:

- **Player characteristics:** age, height, and position.
- **Playing time:** minutes played.
- **Previous injuries:** injury type, days out, matches missed, and injury dates.
- **Season information:** the season in which the player information was recorded.

Data Quality Requirements:
Before modelling, the data needs to be cleaned. This includes:

- Converting numerical and date variables into the correct formats.
- Handling missing values.
- Standardising values such as height and days out.
- Checking inconsistent or duplicate information.
- Checking encoding problems in player names.

Data Leakage:
Only information available before the target season can be used as a predictor. Information about injuries that occur during the target season cannot be used to predict that same season.

Data Structure:
Because a player can have multiple injury records in the same season, the data will need to be organised at player-season level before modelling.

Missing Data:
Missing injury information will be investigated carefully, because an empty injury field does not necessarily mean that the player had no injury.

Data Validation:
After cleaning, the dataset will be checked to ensure that the variables are correctly formatted and that no information from the target season has been used as a predictor.